USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22CSE317]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER -2023

PYTHON FOR IOT

Time: 3 Hours Maximum **Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. Answer All Questions

PART-A **5 X 2=10**

1.	**a.**	Name three primitive data types in Python	L-1	CO1	**2**
	b.	Explain the difference between a 'list' and a 'tuple' in Python	L-2	CO2	**2**
	c.	Describe the concept of a dictionary in Python and provide an example.	L-2	CO3	**2**
	d.	What function is used to initialize the Arduino board and set up the initial settings?	L-1	CO4	**2**
	e.	Name three common operating systems used with the Raspberry Pi.	L-1	CO5	**2**
		PART-B		**6 X 5=30**	
2.	**a.**	Write a python program that checks if a given number is prime or not.	L-3	CO1	**5**
		OR			
	b.	Write a Python script that reverses a number without using built-in functions.	L-3	CO1	**5**
3.	**a.**	Mention the three main control flow structures in Python?	L-2	CO2	**5**
		OR			
	b.	Name three built-in functions in Python and explain their uses.	L-2	CO2	**5**
4.	**a.**	Describe the process of accessing elements by index in a list and a tuple.	L-3	CO2	**5**
		OR			
	b.	Explain how does a dictionary use keys and values to store and retrieve data?	L-3	CO2	**5**
5.	**a.**	Write a Python program that appends elements to a list.	L-3	CO3	**5**
		OR			
	b.	Develop a python script that uses a dictionary to store information about students (e.g., name, age, grade).	L-3	CO3	**5**
6.	**a.**	Write a simple Arduino sketch to control an LED using a push-button.	L-3	CO4	**5**
		OR			
	b.	Develop a program that reads data from a sensor and adjusts the brightness of an LED on the Arduino accordingly.	L-3	CO4	**5**
7.	**a.**	Explain how does the Raspberry Pi utilize Python for programming and interfacing with hardware?	L-2	CO5	**5**
		OR			
	b.	Write a Python program that blinks an LED connected to a GPIO pin on the Raspberry Pi.	L-3	CO5	**5**
		PART-C		**6 X 10=60**	
8.	**a.**	Develop a python code using a Raspberry Pi that reads data from a sensor and displays it using Python.	L-3	CO1	**10**
		OR			
	b.	Consider, You are connecting a PIR (Passive Infrared)	L-3	CO1	**10**

		sensor to the Raspberry Pi. Develop a Python script that detects motion and triggers an action.			
9.	a.	Write a Python code that connects Arduino for performing ON/OFF operation of a LED.	L-3	CO2	10
		OR			
	b.	Compare the functionality between 'analogRead()' and 'digitalRead()' functions in Arduino.	L-4	CO2	10
10.	a.	Evaluate the benefits of using dictionaries over lists for organizing and accessing data in Python.	L-5	CO3	10
		OR			
	b.	Design a python program that combines information from different lists into a dictionary, organizing and categorizing the data.	L-6	CO3	10
11.	a.	Create a file-handling system in Python that involves reading data from one file (csv), processing it, and writing the output to another file (csv).	L-6	CO4	10
		OR			
	b.	Write a Python script that imports a module and uses its functions to solve a specific problem or task.	L-3	CO4	10
12.	a.	Consider a scenario where you need to remove duplicates from a list. Describe the steps and Python methods you would use to accomplish this.	L-4	CO5	10
		OR			
	b.	Explain the process of reading and writing data to a file in Python. Provide examples to demonstrate different modes of file operations and discuss the significance of error handling in file operations.	L-4	CO5	10
13.	a.	Develop a python program to interface Temperature sensor with raspberry pi to monitor temperature.	L-4	CO6	10
		OR			
	b.	Develop a python program to interface Humidity sensor with raspberry pi to monitor the humidity.	L-4	CO6	10